

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Operating Systems

COURSE CODE: CSA0409

COURSE OUTCOME COVERED

CO3: Evaluate memory management mechanisms such as linking, paging, and segmentation to determine their effectiveness for different system requirements

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks: 50

Experiment Questions

- 1.Design and implement a program to simulate the translation of a logical address into a physical address using base and limit registers. The program should accept logical address as input, check for validity, and compute the corresponding physical address. Display appropriate messages for valid and invalid addresses.
- 2.Write a program to demonstrate the paging memory management scheme. The program should divide memory into fixed-size pages and frames, accept page number and offset as input, and compute the corresponding physical address using a page table.
- 3.Develop a program to simulate a page table. The program should allow the user to input page table entries and logical addresses, then map them to physical addresses. Clearly show how page numbers are translated into frame numbers.
- 4.Implement a program to simulate the working of a Translation Lookaside Buffer (TLB). The program should demonstrate how TLB improves memory access time by calculating the effective memory access time with and without TLB.
- 5.Write a program to illustrate memory protection using base and limit registers. The program should verify whether a given logical address lies within the valid range and prevent unauthorized access.



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main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int base, limit, logical, physical;
5
6     printf("Enter base address: ");
7     scanf("%d", &base);
8
9     printf("Enter limit: ");
10    scanf("%d", &limit);
11
12    printf("Enter logical address: ");
13    scanf("%d", &logical);
14
15    if (logical >= 0 && logical < limit) {
16        physical = base + logical;
17        printf("Valid Address\n");
18        printf("Physical Address = %d\n", physical);
19    } else {
20        printf("Invalid Address - Address exceeds limit\n");
21    }
22
23    return 0;
24 }
```



input

```
Enter base address: 1000
Enter limit: 500
Enter logical address: 250
Valid Address
Physical Address = 1250
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

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main.c

```
2
3 int main() {
4     int page, offset, frame, frameSize, physical;
5
6     printf("Enter frame size: ");
7     scanf("%d", &frameSize);
8
9     printf("Enter page number: ");
10    scanf("%d", &page);
11
12    printf("Enter offset: ");
13    scanf("%d", &offset);
14
15    printf("Enter frame number from page table: ");
16    scanf("%d", &frame);
17
18    if (offset >= 0 && offset < frameSize) {
19        physical = frame * frameSize + offset;
20
21        printf("\nPage Number = %d\n", page);
22        printf("Frame Number = %d\n", frame);
23        printf("Offset = %d\n", offset);
24        printf("Physical Address = %d\n", physical);
25    } else {
26        printf("Invalid offset\n");
27    }
28
29    return 0;
30 }
```



input

```
Page Number = 2
Frame Number = 5
Offset = 25
Physical Address = 525
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

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main.c

```
8     scanf("%d", &n);
9
10    printf("Enter frame number for each page:\n");
11
12    for (i = 0; i < n; i++) {
13        printf("Page %d -> Frame: ", i);
14        scanf("%d", &pageTable[i]);
15    }
16
17    printf("\nEnter page number: ");
18    scanf("%d", &page);
19
20    printf("Enter offset: ");
21    scanf("%d", &offset);
22
23    if (page >= 0 && page < n) {
24        frame = pageTable[page];
25
26        physical = frame * 100 + offset;
27
28        printf("\nPage Number = %d\n", page);
29        printf("Frame Number = %d\n", frame);
30        printf("Physical Address = %d\n", physical);
31    } else {
32        printf("Invalid page number\n");
33    }
34
35    return 0;
36 }
```



input

Enter offset: 20

Page Number = 2

Frame Number = 8

Physical Address = 820

...Program finished with exit code 0

Press ENTER to exit console

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main.c

```
25
26     printf("Enter frame size: ");
27     scanf("%d", &frameSize);
28
29     printf("Enter TLB lookup time: ");
30     scanf("%d", &tlbTime);
31
32     printf("Enter memory access time: ");
33     scanf("%d", &memoryTime);
34
35     withoutTLB = 2 * memoryTime;
36
37     if (page == tlbPage) {
38         withTLB = tlbTime + memoryTime;
39
40         printf("\nTLB HIT\n");
41         printf("Frame Number = %d\n", tlbFrame);
42     } else {
43         withTLB = tlbTime + 2 * memoryTime;
44
45         printf("\nTLB MISS\n");
46         printf("Frame Number from Page Table = %d\n", pageTableFrame);
47     }
48
49     printf("Effective Access Time without TLB = %d\n", withoutTLB);
50     printf("Effective Access Time with TLB = %d\n", withTLB);
51
52     return 0;
53 }
```

input

```
TLB HIT
Frame Number = 5
Effective Access Time without TLB = 200
Effective Access Time with TLB = 200
```

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main.c

```
1  #include <stdio.h>
2
3  int main() {
4      int base, limit, address;
5
6      printf("Enter base address: ");
7      scanf("%d", &base);
8
9      printf("Enter limit: ");
10     scanf("%d", &limit);
11
12     printf("Enter logical address: ");
13     scanf("%d", &address);
14
15     if (address >= 0 && address < limit) {
16         printf("\nAccess Allowed\n");
17         printf("Logical Address = %d\n", address);
18         printf("Physical Address = %d\n", base + address);
19     } else {
20         printf("\nAccess Denied\n");
21         printf("Unauthorized memory access!\n");
22     }
23
24     return 0;
25 }
```



input

Enter logical address: 300

Access Allowed

Logical Address = 300

Physical Address = 2300

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...Program finished with exit code 0

Press ENTER to exit console.